**TransformerMPPI** uses a transformer as a learned “first guess” generator for MPPI.

Instead of initializing MPPI from zero or the previous control sequence, the transformer predicts a likely-good mean control sequence from recent states and environment context. MPPI then samples around that better starting point.

They trained the transformer on trajectories generated by regular MPPI, then tested it in simulation on:

- 2D navigation with static/dynamic obstacles
- autonomous racing with obstacles

Main result: **TransformerMPPI usually got lower cost with fewer samples**, especially when sample counts were low. In 2D navigation, it beat MPPI in 8 of 10 low-sample episodes.

It was **not hardware-tested**. The authors explicitly list physical robot deployment as future work.

My take: this is closer to your idea than the clustering paper. It doesn’t reject bad samples directly, but it reduces wasted sampling by starting MPPI near a better region of the control space.